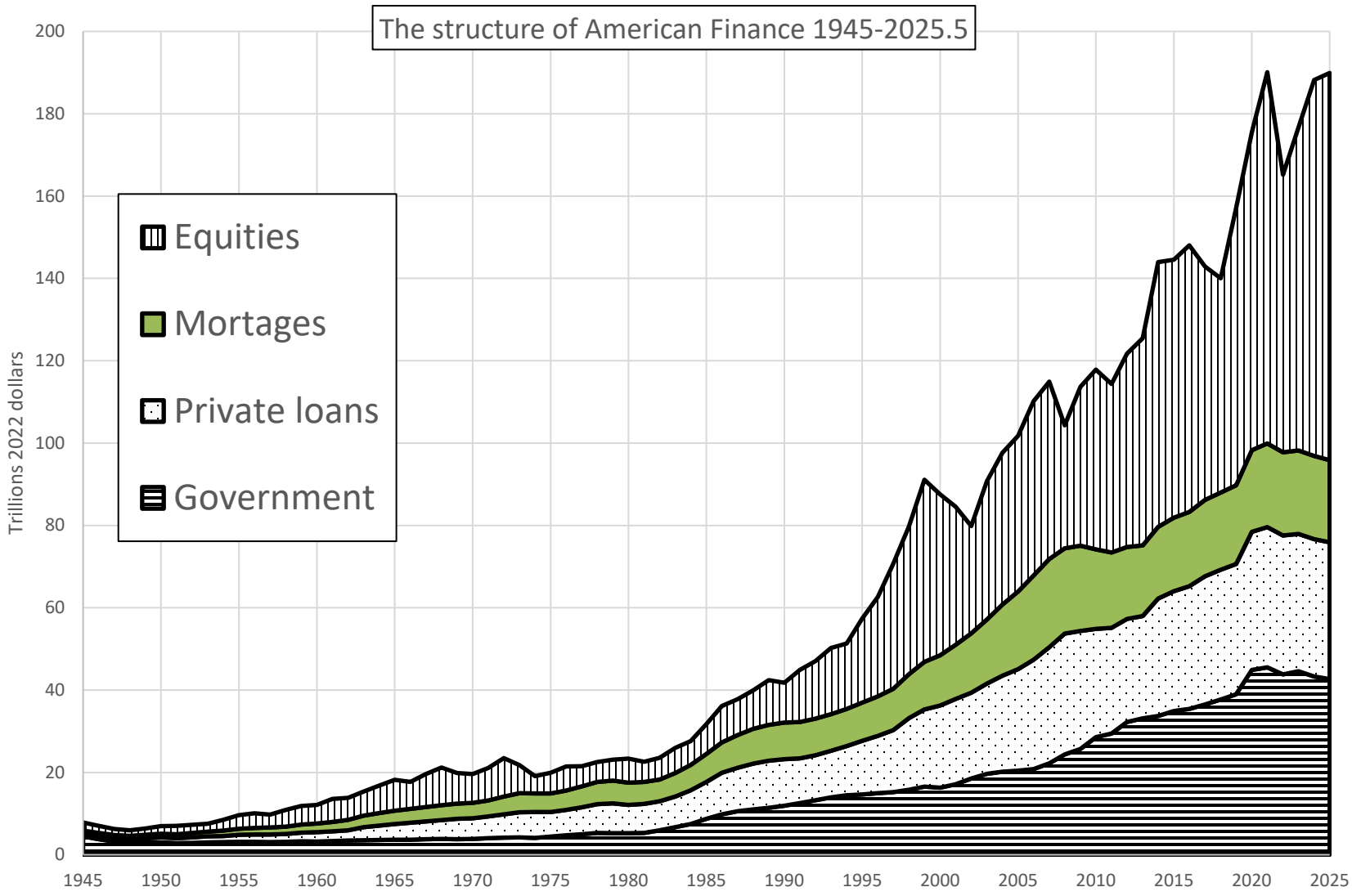
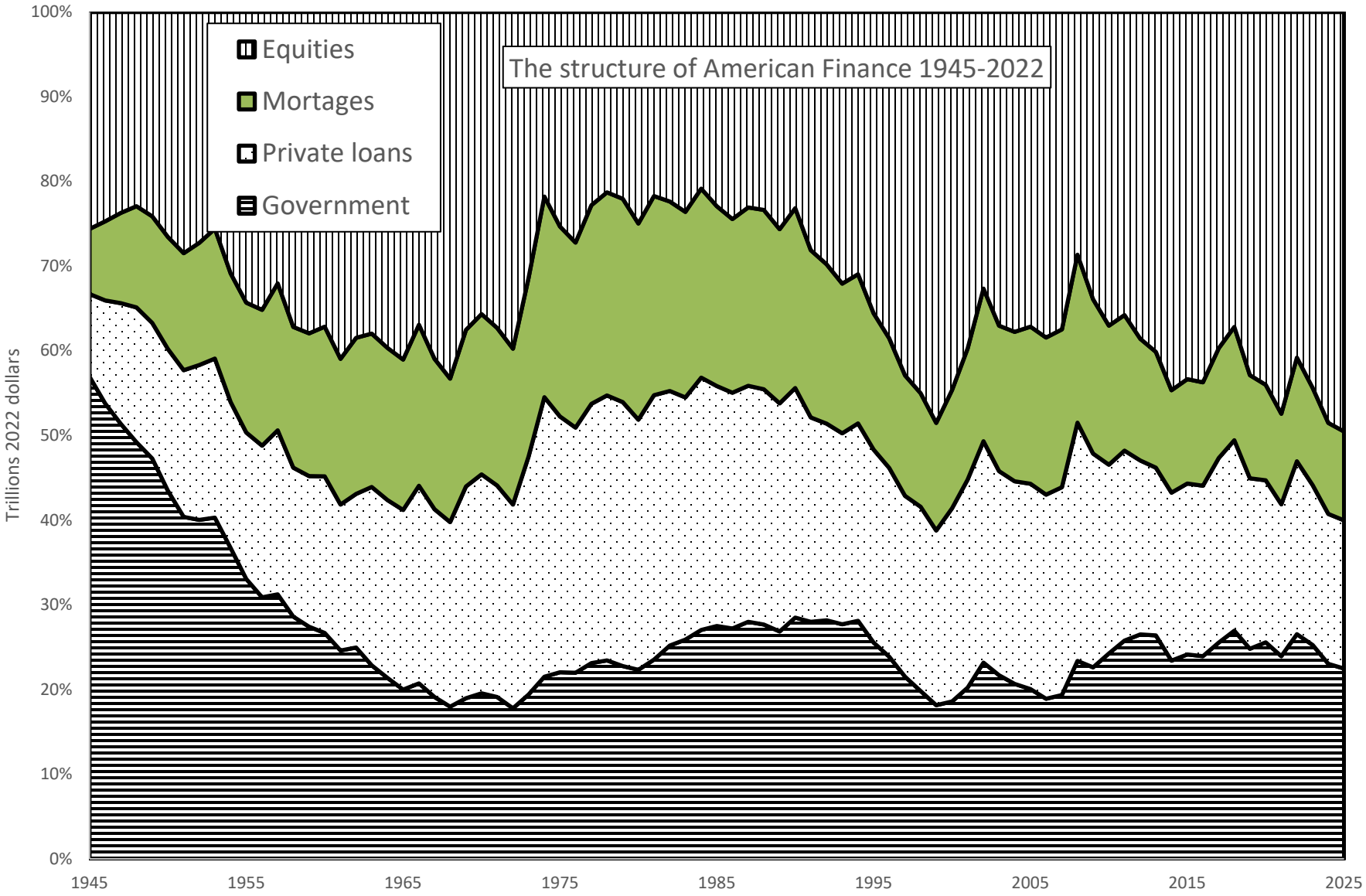
BEM 103

Class 3:
Discounted present value
(time is money)

Demand for capital (supply of claims/assets)
(firms, individuals, government)
Supply of capital (demand for claims/assets)
(firms, individuals, government)
Determining the value of assets
Rates of interest and discount rates
Discounted present value
Rates of return.



Source: (https://www.federalreserve.gov/releases/z1/)


The structure of American Finance 1945-2022
Equities
Mortages
Private loans
Government
Trillions 2022 dollars
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
1945
1955
1965
1975
1985
1995
2005
2015
2025

Price of time

- Start with an individual
  - Consume C now get utility U (C^0)
  - Wait one period to consume C
- Assumption now is better than later

  =>from today's perspective

  U(C^0)> U(C^1) take the ratio $d = \frac{U(C^1)}{U(C^o)}$
- The larger d the closer the utility of consumption today vs tomorrow
- If you invest C from now to next period earn r
  - Now compare U(C^0)> U((1+r)C^1)

Why hold claims?

- Insurance and savings
- Move resources from good states to bad states (ignore for now)
- Move resources from present to future
- How much compare my impatience (d) to my return (r).
- Key idea $d\, vs\, r$
- Claim if r is such that $d = \frac{1}{1+r}$. Then indifferent at the margin between consumption today and consumption tomorrow
- Create an additional claim need to get someone to save more
- => r has to go up.

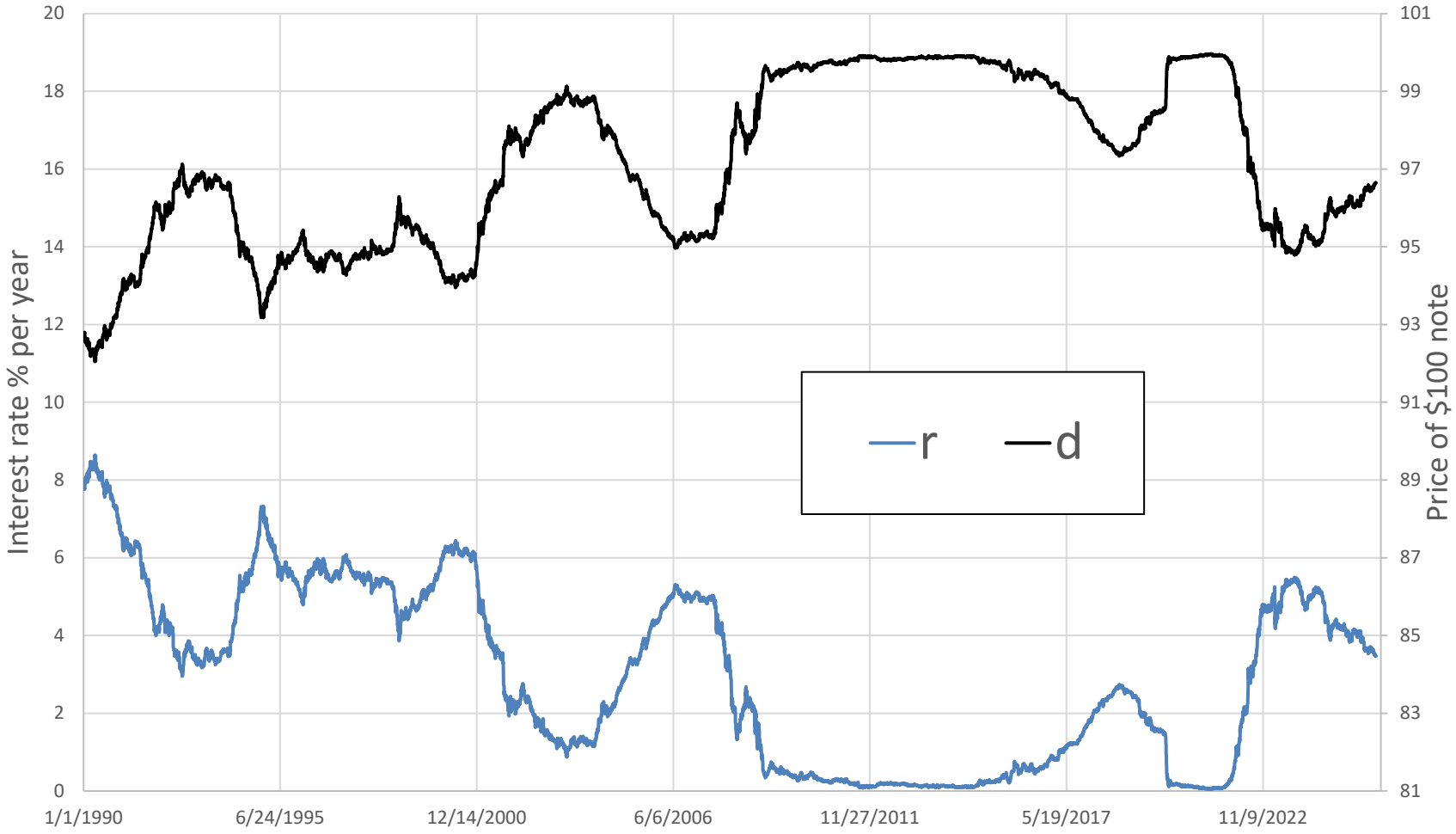
The price of a 1 year no coupon riskless bond

- Bond pay 100 in 1 period
- Interest rate is r
- What is price of bond

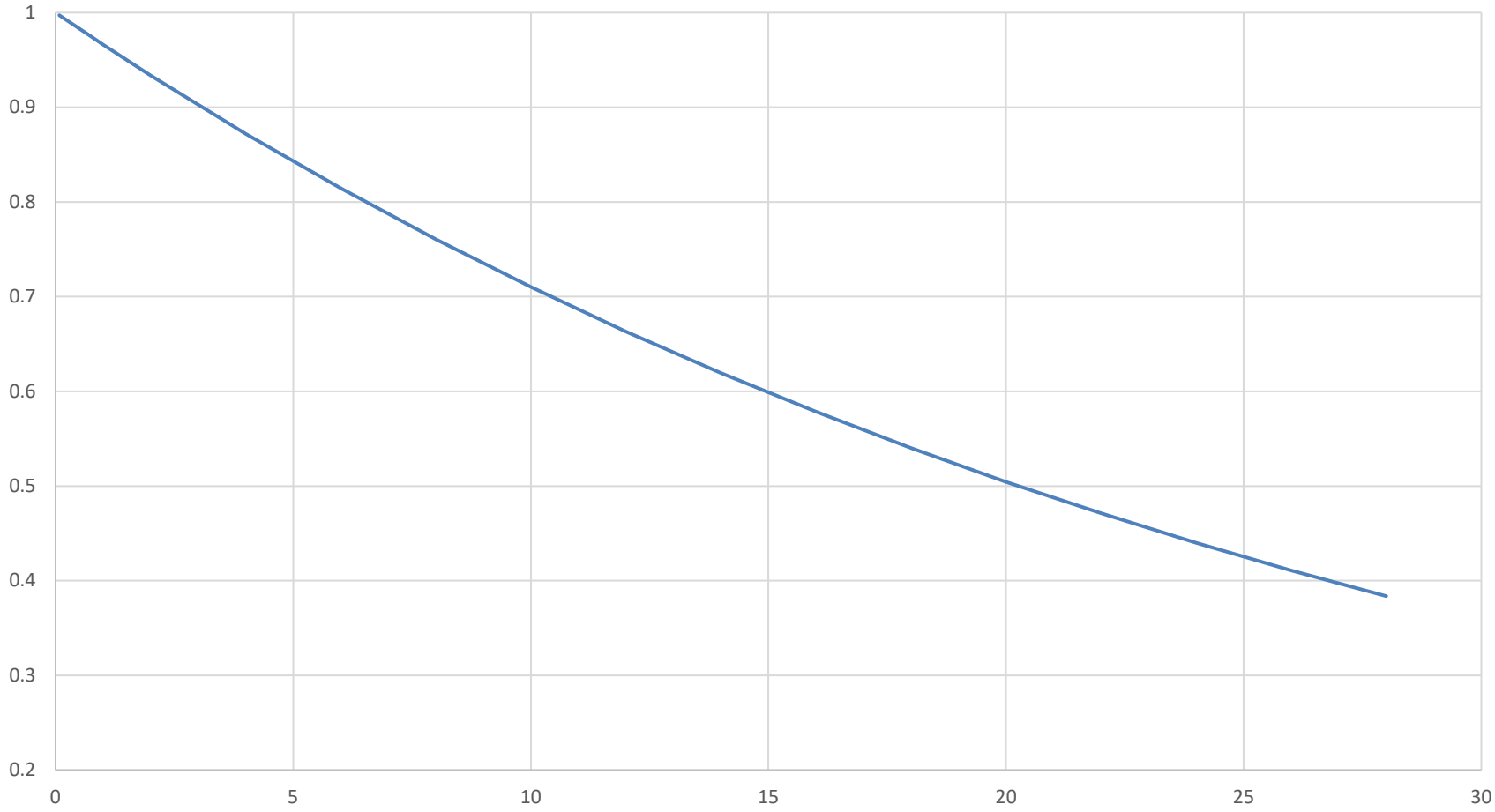
1. Pay p get 100 in one period
2. pr save p and get p(1+r) in one period.

- No arbitrage $\Rightarrow p(1+r) = 100 \Rightarrow \ p = \frac{100}{1+r}$
- The discount on the bond is $d = \frac{1}{1+r}$

r and d over time



In the riskless world interest compounds so discount increases with time



Focus on the problem for a firm

- **Acceleration**
- **Firms have two sources of investment resources**
  - Retained earnings,
  - Issuing new claims
- **If investment is limited to retained earnings it is constrained by profits.**
  - Problems for firms with indivisible investment (it may take such a long time to accumulate the needed resources, investment never takes place)
  - Problem for firms where market share matters for long term strategic growth (imitation is easy but customers are loyal)
  - You get pro-cyclical investment and that may not maximize profits—in particular in industries where there are time-to-build delays.

Example aviation: it takes a decade to design and test a new jet liner. And you want to be ready when demand is forecast to be large.

Project Evaluation

- **Project: An investment opportunity**
  - New firm?
  - Firm starts a new investment process?
- **Definition: Project is vector of investments c and a vector of incomes Y over T periods**
- $C=\{c_1, c_2,..,c_t \ldots c_T\}$ some may be zero
- $Y=\{y_1, y_2,..,y_t \ldots y_T\}$ some may be zero
- let $x_t = y_t - c_t$ $X=\{x_1, x_2,..,x_t \ldots x_T\}$.
- X is the vector of cash flows
  - Each x_t can be positive or negative
  - So now how do we figure what X is worth?

Deciding value

- $X=\{x_1, x_2,..,x_t \ldots x_T\}$.

Principle: Cash flow soon is better than cash flow later.

Idea: Suppose you were going to borrow V to buy X and pay off your loan with the cash flows

How much would you be able to borrow?

The one period version

- Interest rate is r, borrow V to buy it
- $X = \{x_1\}$ $\qquad \pi = x_1 - (1+r)V$

$$\pi = 0 \Leftrightarrow V = \frac{x_1}{1+r} \Leftrightarrow V = dx_1$$

- $p(X) - V$=net present value
  - The value of the project taking into account opportunity cost of capital
  - Buy X if NPV>0. Here individual d does not matter. r is market rate.
- Axiom 1: markets are competitive

$$\pi = 0 \Leftrightarrow dx_1 = p(x_1) = p(X)$$

More periods

- $X = \{x_1, x_2\}$ $\quad V = d(x_1 + dx_2) = dx_1 + d^2 x_2$

$$V = \frac{x_1}{1+r} + \frac{x_2}{(1+r)^2}$$

- Double discounting (or interest for two periods with compounding)
- If borrow V, can just pay it off with my cash flow so $\pi = 0$
- Notice that we are using additivity axiom here.

Discounted present value of a cash flow X

- X={x_1, x_2,..,x_t …x_T} is $V = \sum_1^T \frac{x_t}{(1+r)^t} = \sum_1^T d^t x_t$
- For each cash flow (x_t) you can borrow less the later it arrives because you will have to pay more accumulated interest

Deciding value

Discounted present value

Core concept

Notice

(1) if different people (places) have different r, they have different discounted present values

(2) the same project presented in a period when interest rates are low is more attractive than when interest rates are high. (Value vs Growth investing)

(3) this is one reason central banks matter (they play an important role in setting r).

- Interest rate and discount rate $d = \frac{1}{1+r}$

$V = \sum_1^T \frac{x_t}{(1+r)^t} \quad = \sum_1^T x_t \, d^t$

N projects

- Let the firm contemplate n projects
  - The longest lived ends at time T
  - For each project (i) we can define
  - $X^i=\{x^i_1, x^i_2,..,x^i_t \ldots x^i_T\}$.
  - So we can compute NPV for each.
- Which one should we undertake?
  - Are early positive x^i good or bad? Are large negative followed by large positive x^i
  - One needs a mechanism for inter-temporal accounting

Choosing among projects

Rules

- If capital is widely available, do all positive NPV projects (max total profits not the rate of return)
- If capital constrained do those with the largest NPV subject to budget constraint
- The firm should ask:
  - How many of these projects are profitable enough that the market would want to fund them?
- But that means we must measure value

Cost benefit ratio

- Related concept often used in civil engineering

- Project has revenues X={y_1, y_2,..,y_t …y_T} and costs C={c_1, c_2,..,c_t …c_T}.

- The cost benefit ratio is : $CB = \frac{\sum_1^T y_t d^t}{\sum_1^T c_t d^t}$

- Recall $NPV = \sum_1^T (y_t - c_t) d^t = \sum_1^T (y_t) d^t - \sum_1^T (c_t) d^t$

- $\sum_1^T (y_t) d^t - \sum_1^T (c_t) d^t = 0 \Leftrightarrow \frac{\sum_1^T y_t d^t}{\sum_1^T c_t d^t} = 1$

- So NPV>0 iff CB>1

Internal rate of return

- Internal rate of return find r_i such that

$$V = \sum_1^T \frac{y_t - ct}{(1+r_i)^t} = 0$$

  - Problem generically T-1 roots (some imaginary)
    - Pick the smallest real root.
- Let r_M be the market rate of interest.
  - Evaluate CB and NPV using $d = \frac{1}{1+r_M}$
  - In general, if $r_i > r_M$ CB>1 and NPV>0
  - Problem 2:late negative payoff negative cause pbs (because high r_i imply you do not care much about the future)

The problem of value (wrap up)

- To establish value need to deal with time
- NPV, IRR, and CB are essentially all the same for decision making
- For asset valuation. We start with NPV.
  - Satisfies basic axioms of finance
  - It does not matter what the asset is.
    - What matters is its cash flow to the investor
  - Answers what is the most anyone would pay for this?

Class 05: Pricing Riskless Bonds

- Net Present value, again;
- Fed policy and inflation;
- Expectations and the yield curve;
- Implication for assets with risk;
- Application: QE2 and the unwinding of the Fed Position